

Sarthak Shah

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SUMMARY

Senior Software Engineer with 8+ years building backend microservices, RESTful APIs, and cloud-native platforms across financial services, aerospace, medical devices, and IoT. Expert in Python (FastAPI, Django, Flask) with deep PostgreSQL, Redis, and DynamoDB experience, including schema design, query optimization, and caching strategy. Strong system design foundation in scalability, fault tolerance, and performance, with hands-on ownership of AWS deployments, Docker and Kubernetes workloads, CI/CD pipelines, and production web servers (Nginx, Caddy, Gunicorn). Comfortable working directly with business and IT stakeholders to translate requirements into production services, and experienced in observability and log-driven monitoring using Grafana Loki and Splunk. Currently expanding into Golang (Fiber) and AI application development.

SKILLS

Programming: Python, Go, SQL, C/C++ (embedded)

Backend Frameworks: FastAPI, Django (DRF), Flask, Fiber (Go)

API Development: RESTful APIs, OpenAPI/Swagger, Webhooks, MQTT, JSON-RPC, Modbus

Architecture & Design: Microservices, system design (scalability, fault tolerance, performance), caching strategies, async and multi-threaded processing

Databases: PostgreSQL, Redis, DynamoDB, Django ORM, pgAdmin

Cloud & DevOps: AWS (EC2, S3, Lambda, CodePipeline), Docker, Kubernetes, Jenkins, GitHub Actions, Git, Bitbucket

Web Servers: Nginx, Caddy, Gunicorn

Monitoring & Observability: Grafana Loki, Splunk, structured logging

Data Workflow: Apache PySpark, Iceberg, Airflow, JupyterHub

Testing & Automation: Pytest, Unittest, Selenium, Robot Framework, Postman

CURRENTLY LEARNING

AI App Dev: LangChain, LangGraph, Snowflake-arctic-embed, Qdrant (vector search, embeddings, agentic workflows)

WORK EXPERIENCE

Senior Software Developer

Aug 2025 – Present

Libellule Monde, Toronto (remote)

- Manage PostgreSQL and DynamoDB databases for an aircraft data and analytics platform, continually optimizing query efficiency and storage strategy using Django ORM and pgAdmin.
- Implemented Redis as a caching layer to cut database round-trip times and serve pre-indexed search suggestions, significantly improving response latency.
- Built a Microsoft Power Automate workflow integrated via Webhook API to deliver automated daily email and Microsoft Teams updates to each user.
- Own backend service deployments on AWS EC2, maintaining Docker-based environments and web server configuration to keep production and development parity.

Senior Software Engineer

Apr 2025 – Aug 2025

First Derivative, ON (Client engagement: Financial Services / Capital Markets)

- Developed Python scripts to extract user activity from trading application logs across NAS file systems, databases, and Splunk; used regex and schedulers to transform logs into structured JSON capturing timestamped queries, user actions, and originating screens for Risk team anomaly detection.
- Partnered with IT system owners to enhance application logging strategies, enabling granular tracking of trader interactions and improving surveillance precision.
- Deployed region-specific log access jobs across distributed environments via enterprise frameworks; monitored job status and resource impact using Grafana Loki and Splunk.
- Designed a Qualtrics-based technical onboarding survey with embedded data fields and branching logic to capture application-specific configurations from IT system owners, centralizing monitoring enablement across teams.

Senior Software Engineer

Oct 2023 – Apr 2025

Libellule Monde, Toronto (remote)

- Led design and development of a Django-based smart configurator platform integrating OpenCV and image annotation tools to analyze aircraft exterior regions, enabling accurate estimation of time, resources, and cost for surface treatments such as decals and paint masks.
- Built and maintain a FastAPI RESTful service for aircraft parts label generation, automating enterprise-specific labels for parts, stickers, and plates with efficient search across a large-scale part number database.
- Designed and tuned PostgreSQL and DynamoDB data models for the aircraft analytics platform, profiling slow queries and reshaping indexes to meet performance benchmarks as datasets grew.
- Integrated a YOLOv7 detection model trained with Roboflow into the computer vision pipeline, improving object detection accuracy and throughput in near real-time processing.
- Oversaw Gunicorn and Caddy web servers on AWS EC2 for production and development, troubleshooting and tuning configurations for performance and scalability.

- Containerized Django applications with Docker, ensuring consistent development environments across the team.

Senior Software Engineer

Feb 2022 – Oct 2023

Bacancy Technology

- Developed FastAPI REST endpoints leveraging AWS S3, MQTT, and Lambda to integrate Philips Biotel medical devices with the client portal, ensuring secure storage and real-time processing of device data.
- Architected Django web applications using Django ORM, authentication and authorization mechanisms, custom template tags, and DRF; hosted on EC2 with Nginx and Gunicorn.
- Containerized Flask applications with Docker and orchestrated services with Kubernetes, improving scalability and fault tolerance.
- Maintained and deployed a Flask-based application on AWS EC2 behind Gunicorn, handling medium traffic loads with monitored uptime and rollback-safe releases.

Embedded Software Engineer

Oct 2020 – Feb 2022

Xylem Inc.

- Led development of a Python-based embedded application for water controllers; used CPython to convert Python code into C modules for flashing onto microcontrollers.
- Built real-time applications with Python signal, threading, and scheduler libraries using thread pool and process pool models.
- Integrated Modbus, MQTT, and JSON-RPC protocols to enable reliable sensor-to-controller communication across distributed field devices.

Verification & Validation Engineer

Feb 2020 – Aug 2020

A&W Engineering Works

- Developed verification test cases for ultrasound medical hardware in Python and C++, reducing software defects by 95% and testing time by 30%.
- Built computation and reporting utilities with NumPy, pandas, and matplotlib to analyze and plot signal data for hardware validation cycles.

Software Engineer

Aug 2018 – Jan 2020

L&T Technology Services

- Built a Robot Framework solution in Python for mobile, tablet, and OTT app testing, reducing testing time by 40% and increasing test coverage by 25%.
- Developed and tested end-to-end REST API scenarios using the requests and json modules alongside Postman for validation.
- Automated technical datasheet mapping and validation (data cleansing, correlation, extraction) using pandas, NumPy, and Openpyxl.

EDUCATION

Post Graduation, Full Stack Software Development

Aug 2022 – Apr 2024

Lambton College, Toronto, Canada

- Advanced Python and Go, relational and non-relational database management, cloud computing and containerization.

B.Tech, Electrical and Electronics

May 2012 – Apr 2016

Indus University, India

- Embedded systems design, wireless communication, digital and power electronics.

AWARDS

- Two Spot Awards, L&T Technology Services: Python-based data validation; OCR-based performance testing for OTT applications.
- Spot Award, Bacancy Technology: quick adaptability with new projects.